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Very Important Publication

An Allene Approach to Tetrahydrofuran-Fused Bicyclo[2.2.2]oct-2-enes

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Abstract: An efficient access to tetrahydrofuran-fused bicyclo[2.2.2]oct-2-enes via palladium-catalyzed cyclization of oxygen-tethered 2,7-alkadiynylic carbonates with 2-(buta-2,3-dienyl)malononitrile has been developed. Perfect chiral induction was observed when a substituent was introduced to the propargylic position affording the optically active polycyclic products of both enantiomers. Control experiments showed that the oxygen tether of 2,7-alkadiynylic carbonates and the two cyano groups in allene are indispensible for this transformation.

Keywords: palladium; polycycle; allene; 2,7-alkadiynylic carbonates

During the last 10 to 15 years much attention has been paid to transition metal-catalyzed cyclization reactions of different types of allenes.[1,2] In this area, we have studied the cyclization of allenes bearing a nucleophilic functionality in the presence of 2,7-alkadiynyl carbonates affording tricyclic products **A**,[3a] **B**,[3b] or **C**.[4] Interestingly when 2-(buta-2,3-dienyl)malononitrile was used instead of allenoic acids, allenyl amides, or allenyl malonate or bis(phenylsulfonyl)methane, tetrahydrofuran-fused bicycle[2.2.2]oct-2-enes **3** was formed unexpectedly. The core structure **D**, a bicyclo[2.2.2]oct-2-ene fused with a tetrahydrofuran ring, is also a structural element of Tokinolide B and Diligustilide, which have been isolated from the extracts of rhizome of *L. porter*. They show significant inhibitory effects against inflammation[5a] and human cancer cells.[5b] In addition, a more complex and potential chemical, chanxiongnolide B,[5c] was also isolated. From a pharmacological point of view, an efficient access to such bicyclo[2.2.2]oct-2-ene fused with a tetrahydrofuran ring **D** may lead to the discovery of new lead compounds. However, the existing synthesis of such bridged core structure **D** is still very limited.[6] In this paper, we wish to report this efficient palladium-catalyzed entry to tetrahydrofuran-fused bicyclo[2.2.2]oct-2-enes **3** from the readily available oxygen-tethered 2,7-alkadiynylic carbonates with 2-(buta-2,3-dienyl)malononitrile (Scheme 1).

Initially, when 2,7-alkadiynylic carbonate (**1a**) was subjected to react with 2-(buta-2,3-dienyl)malononitrile (**2a**) under the catalysis of $Pd(OAc)_2$/Gorlos-Phos·HBF_4[7a] using K_2CO_3 as the base at 50 °C for 7 h in CH_3CN, quite unexpectedly, a new product (**3a**) containing a bicyclo[2.2.2]oct-2-ene fused with a tetrahydrofuran ring was formed (Table 1, entry 3). Its structure was established by X-ray diffraction analysis (Figure 1).[8]

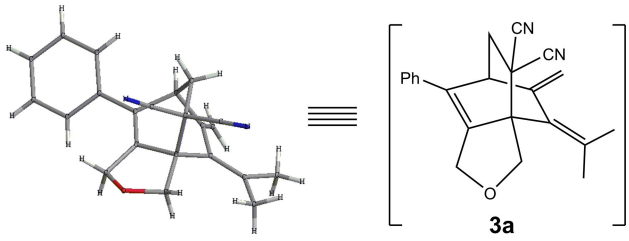
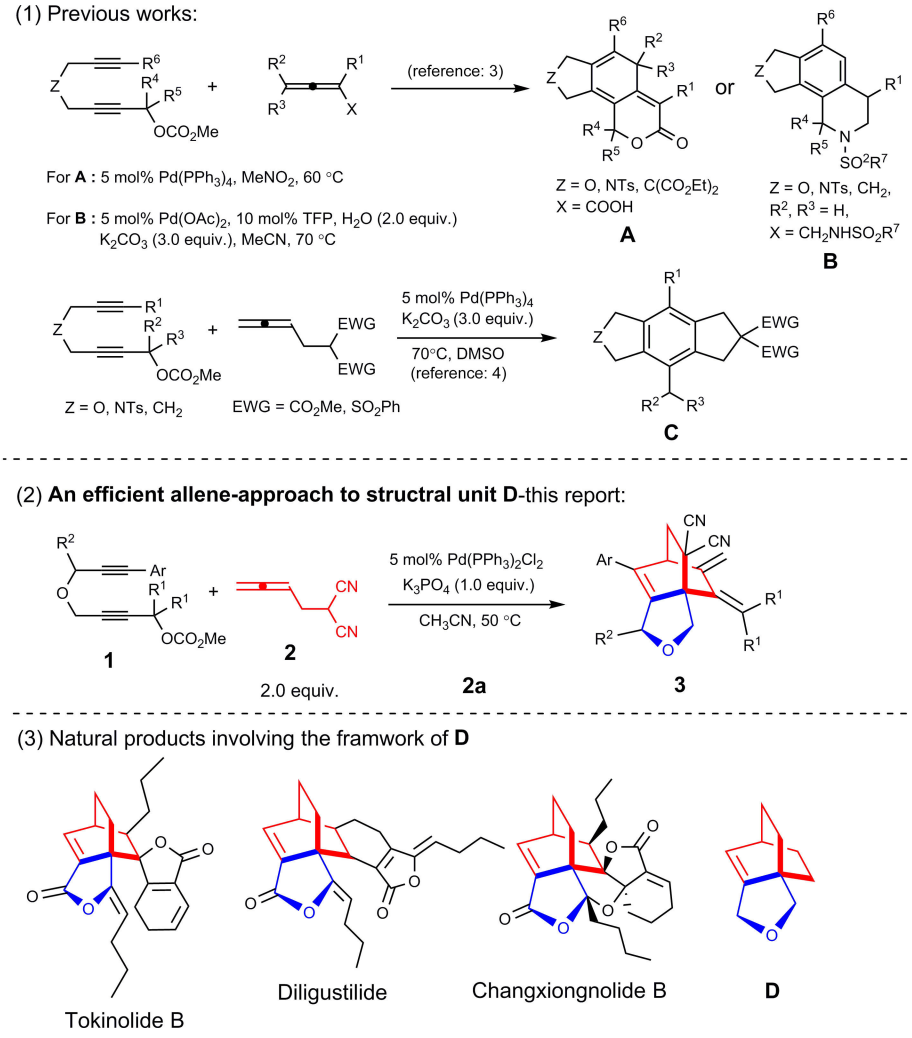


Figure 1. The ORTEP Representation of **3a**.

Due to the above-mentioned importance of such a core structure **D** in Scheme 1, we set out to optimize the reaction conditions. Initially, various phosphorus ligands and palladium resources were examined. When submitting $Pd(OAc)_2$ as catalyst with LB-Phos·HBF_4[7b] or Zheda-Phos·HBF_4[7c] as ligand, the reaction failed (Table 1, entries 1 and 2). By using Gorlos-Phos·HBF_4, the bridged polycycle **3a** was formed in a yield of 12% in addition to 4% of **4a**[3] and 1% of **5a**[4] with recovery of **1a** in 57% yield

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Scheme 1. The Previous Works and the Synthesis of Tetrahydrofuran-fused Bicyclo[2.2.2]oct-2-enes.

(Table 1, entry 3). With DBA (dibenzylideneacetone) as a ligand very limited improvement was observed (Table 1, entry 4). Interestingly, PPh_3 gave a moderate yield of **3a** of 50%, however, more **4a** and **5a** were produced too (Table 1, entry 5). $Pd(PPh_3)_4$ showed a slightly lower efficiency but a better selectivity (Table 1, entry 6). $Pd(PPh_3)_2Cl_2$ gave a better yield of 51% for **3a** and the reaction selectivity was improved further (Table 1, entry 7). Some other experimental parameters were also screened and found that the molar ratio of reactants has a great influence on the reaction: 57% yield of **3a** and 6% of **4a** were observed with a molar ratio of **1a**/**2a** being 1/2 (Table 1, entry 8). With K_3PO_4 as a base, the formation of **4a** was surpressed to 2% with 61% of **3a** being formed (Table 1, entry 10). Thus, **2a** (2.0 equiv), $Pd(PPh_3)_2Cl_2$ (5 mol%), K_3PO_4 (1.0 equiv) in CH_3CN at 50 °C have been defined as the standard reaction conditions for further study.

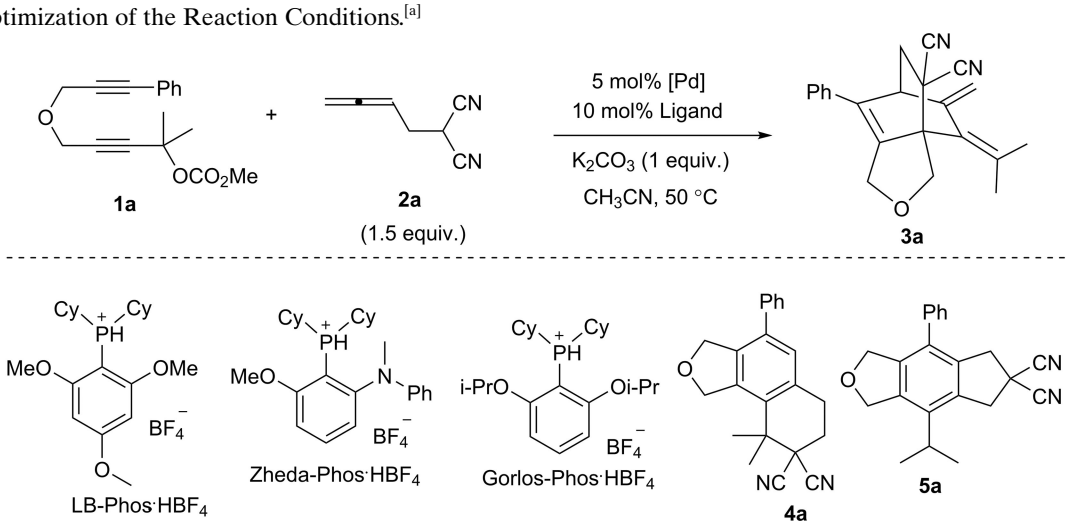
With the optimized conditions in hand, the substrate scope of the reaction was examined. The reaction proceeded well on a gram-scale (Table 2, entry 3). The electronic effect of the Ar group has a very limited impact on the efficiency of the transformation: A carbon-halogen bond on the benzene ring is also tolerated (Table 2, entries 5 and 6), which makes it possible for further functionalization; thienyl is also compatible (Table 2, entry 7). R^1 may be two alkyl groups or a cyclic alkyl group (Table 2, entries 9~11).

When non-symmetric 2,7-alkadiynylic carbonate **1k** was reacted with 2-(buta-2,3-dien-1-yl)malononitrile, (*Z*)-**3k** and (*E*)-**3k** were produced in 51% yield with a *Z*/*E* ratio of 2.3:1. The configurations of (*Z*)-**3k** and (*E*)-**3k** were tentatively assigned by NOE analysis (see eq. 1).

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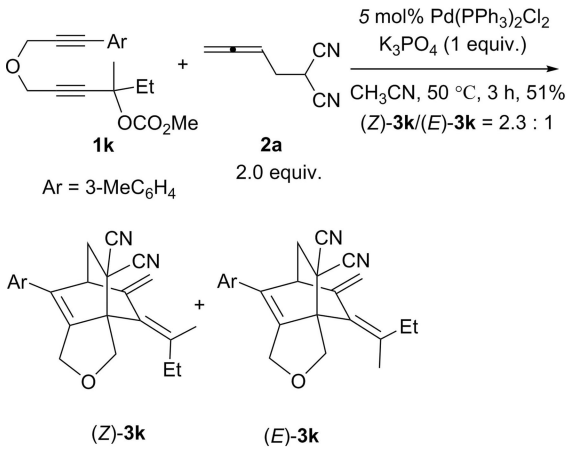
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Table 1. Optimization of the Reaction Conditions.[a]



Entry	Catalyst	Ligand	t/h	NMR yield (%)[b] **3a/4a/5a**	Recovery of **1a** (%)[b]
1[c]	$Pd(OAc)_2$	LB-Phos·HBF_4	7	–/–/–	82
2[c]	$Pd(OAc)_2$	Zheda-Phos·HBF_4	7	–/–/–	62
3[c]	$Pd(OAc)_2$	Gorlos-Phos·HBF_4	7	12/4/1	57
4	$Pd(OAc)_2$	DBA	21	6/7/–	56
5	$Pd(OAc)_2$	PPh_3	4	50/16/3	–
6	$Pd(PPh_3)_4$	–	3	40/8/1	–
7	$Pd(PPh_3)_2Cl_2$	–	1.5	51/9/1	–
8[d]	$Pd(PPh_3)_2Cl_2$	–	1.5	57/6/2	–
9[e]	$Pd(PPh_3)_2Cl_2$	–	3	30/24/3	–
10[d,f]	$Pd(PPh_3)_2Cl_2$	–	1.5	61/2/1	–

[a] Reaction conditions: **1a** (0.3 mmol), **2a** (0.45 mmol), catalyst (0.015 mmol), Ligand (0.03 mmol), K_2CO_3 (0.3 mmol) in CH_3CN (3 mL) at 50 °C, and monitored by TLC.
[b] Determined by ^1H NMR analysis using mesitylene as internal strandard.
[c] Two equivalents of K_2CO_3 were used.
[d] Molar ratio of **1a**/**2a** = 1:2.
[e] Molar ratio of **1a**/**2a** = 1:2.2.
[f] K_2CO_3 was replaced by K_3PO_4.



(1)

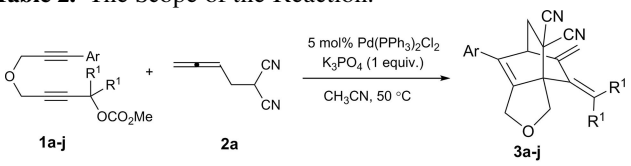
Furthermore, it is exciting to observe that the reactions of racemic compounds **1l** and **1m** afforded **3l** and **3m** as one diastereomer exclusively, respectively. Thus, the reactions of optically active (*R*)-**1l** and its enantiomer (*S*)-**1l** were tried and the enantiomers, (1*R*,3*R*,7*S*)-**3l** and (1*S*,3*S*,7*R*)-**3l**, were formed, exclusively, demonstrating an interesting highly efficient chiral induction to control all the remaining absolute configurations (Scheme 2). Their absolute configurations were established by X-ray diffraction analysis.[9]

To our surprise, the cascade reaction towards **3** failed when 2,7-alkadiynylic carbonate **1** was tethered with CH_2 or NTs, indicating a unique role of the oxygen atom. Unfortunately, the reaction of sulfur-tethered 2,7-alkadiynylic carbonate also did not afford the expected tricyclic product either, which may be caused by the poison of the palladium catalyst by the sulfur atom. In addition, when one CN group in **2a** was replaced with a methoxycarbonyl group (**2b**), no bridged polycyclic product was formed, indicting the

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Table 2. The Scope of the Reaction.[a]



Entry	**1** Ar	R^1, R^1	t/h	Yield of **3** (%)[b,c]
1	Ph	Me, Me (**1a**)	2	64 (51) (**3a**)
2	4-MeC_6H_4	Me, Me (**1b**)	2.5	67 (53) (**3b**)
3[d]	4-MeC_6H_4	Me, Me (**1b**)	2.5	68 (55) (**3b**)
4	4-$MeOC_6H_4$	Me, Me (**1c**)	2	63 (55) (**3c**)
5	4-FC_6H_4	Me, Me (**1d**)	3	64 (50) (**3d**)
6	4-ClC_6H_4	Me, Me (**1e**)	3.5	57 (42) (**3e**)
7	3-thienyl	Me, Me (**1f**)	3	65 (52) (**3f**)
8	Ph	Et, Et (**1g**)	2.5	66 (54) (**3g**)
9	Ph	$(CH_2)_4$ (**1h**)	6	67 (56) (**3h**)
10	2-MeC_6H_4	$(CH_2)_4$ (**1i**)	7	54 (43) (**3i**)
11	Ph	$(CH_2)_6$ (**1j**)	17	53 (36) (**3j**)

[a] Reaction conditions: **1a** (1.0 mmol), **2a** (2.0 mmol), $Pd(PPh_3)_2Cl_2$ (0.05 mmol), K_3PO_4 (1.0 mmol) in CH_3CN (10 mL) at 50 °C, and monitored by TLC.
[b] NMR yield determined by ^1H NMR analysis using mesitylene as internal strandard.
[c] The numbers in the parenthesis were isolated yields.
[d] The reaction was conducted on 5 mmol scale.

two cyano groups are indispensible for this transformation (Scheme 3).

On the basis of above results, a plausible mechanism was tentatively proposed as shown in Scheme 4 by taking the reaction of **1a** with **2a** as the example. Oxidative addition of **1a** with the catalytically active species Pd(0) would give the allenylpalladium intermediate **IN-1**,[10,11] which undergoes intramolecular insertion of the triple bond to generate the alkenylpalladium species **IN-2**.[11] The species **IN-2** undergoes highly regioselective intermolecular carbopalladation of the internal C=C bond of the allene moiety in **2a** to generate a new stereodefined alkenylpalladium intermediate **IN-3** due to the side-on coordination of C≡N with palladium (**Path a**).[12] Then intramolecular carbopalladation of allene would form a new allylpalladium intermediate **IN-4** or **IN-4′**. In presence of a base, the key intermediate **IN-4** or **IN-4′** could undergo intramolecular nucleophilic attack forming bridged tricyclic product **3a** and regenerating the catalytically active Pd(0). The intermediate **IN-2** may also undergo intermolecular carbopalladation of the terminal C=C bond of the allene moiety in **2a** to generate a stereodefined alkenylpalladium intermediate **IN-5** with palladium coordinated with C≡N in side-on mode (**Path b**).[3b] Then intramolecular carbopalladation of allene and intramolecular nucleophilic attack would form tricyclic intermediate **IN-7** and regenerate the catalytically active Pd(0). After aromatization, the side-product **4a** was delivered. Compared with path b, the coordination of the two CN groups with Pd may have determined the regioselectivity of the insertion reaction forming intermediate **IN-3**.

In summary, we have developed a highly efficient approach to tetrahydrofuran-fused bicyclo[2.2.2]oct-2-enes from the readily available oxygen-tethered 2,7-alkadiynylic carbonates and 2-(buta-2,3-dien-1-yl)malononitrile under the catalysis of palladium. This transformation features mild conditions and efficiency to such a complicated skeleton. Optically active bridged products also can be yielded with perfect chiral induction. Control experiments show that the oxygen tether of 2,7-alkadiynylic carbonates and the two cyano groups of allenes are critical for this transformation although the role of these functionalities is not clear to us. The further utilization of the transformation and study on the role of each reactant are going on in our research group.

Experimental Section

General Experimental Methods

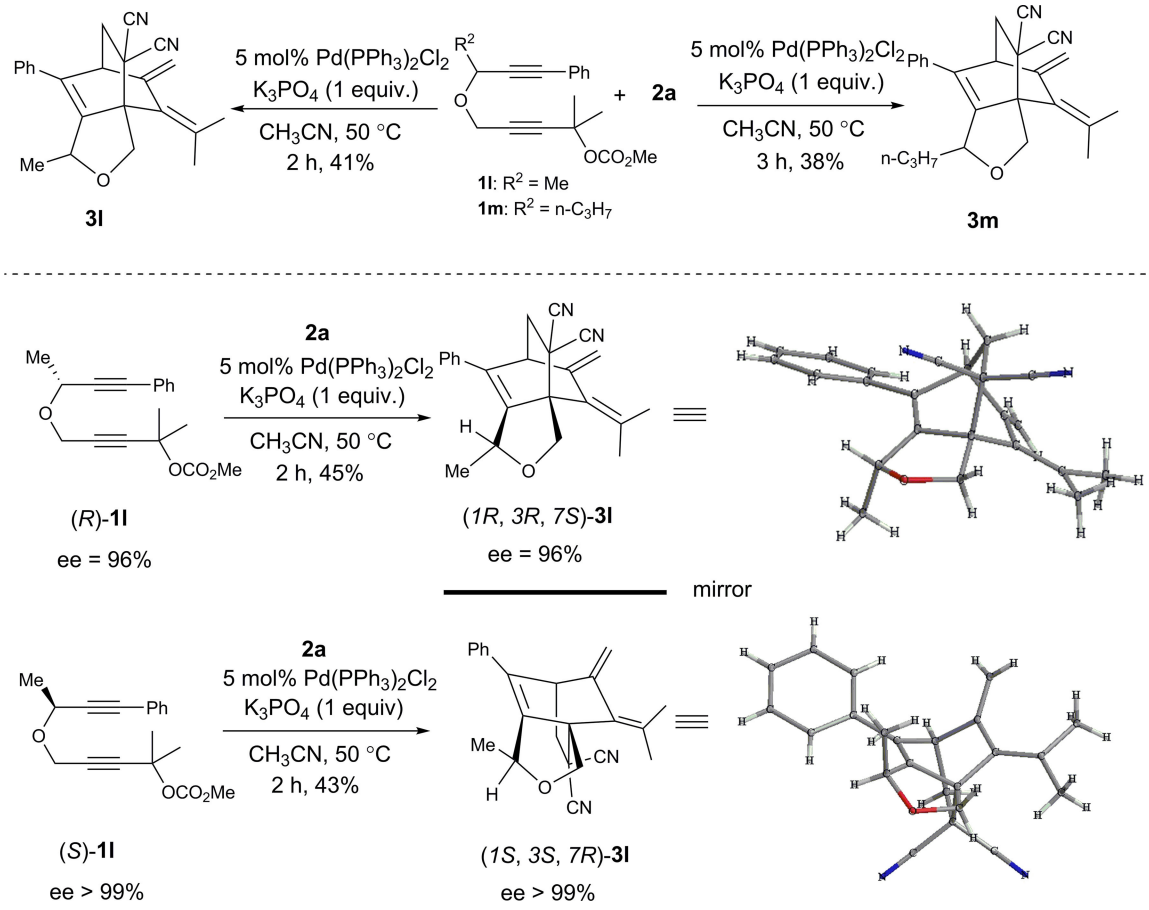
$Pd(PPh_3)_2Cl_2$ was purchased from Energy Chemical. K_3PO_4 was purchased from Shanghai Titan Scientific Co., Ltd. CH_3CN used was dried over CaH_2 under reflux conditions and distilled just before use. The synthesis of starting material 2,7-alkadiynylic carbonates **1a–1c**, **1e**, **1h–1i**, and **1l–1n** were prepared according to the literature procedures.[4] The boiling range of the petroleum ether was 30–60 °C unless noted otherwise. Other commercially available chemicals were purchased and used without additional purification unless noted otherwise. All ^1H NMR experiments were measured with tetramethylsilane (0 ppm) or the signal of $CHCl_3$ (7.26 ppm) in $CDCl_3$ as the internal reference; ^{13}C NMR experiments were measured in relative to the signal of $CDCl_3$ (77.0 ppm); ^{19}F NMR experiments were measured in relative to the signal of $CFCl_3$ (0 ppm) in $CDCl_3$. Infrared spectra were recorded from the film of pure samples on sodium chloride plates for liquid. Mass and HRMS spectra were carried out in EI mode. Elemental analyses were carried out by Elementar Vario MICRO cube. Thin layer chromatography was performed on pre-coated glassback plates and visualized with UV light at 254 nm. Flash column chromatography was performed on silica gel.

Typical Procedure for the Synthesis of Tetrahydrofuran-fused Bicyclo[2.2.2]oct-2-enes 3

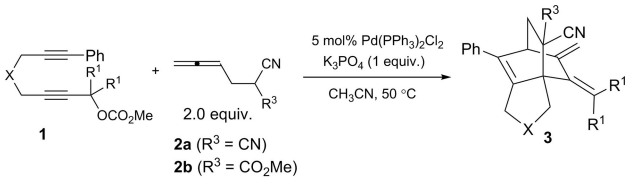
Synthesis of **3a**: To a Schlenk tube containing K_3PO_4 (213.0 mg, 1.0 mmol) dried with heating gun under vacuum and filled with nitrogen, $Pd(PPh_3)_2Cl_2$ (34.9 mg, 0.05 mmol), **1a** (286.4 mg, 1.0 mmol), and CH_3CN (3.0 mL) were added, then **2a** (236.3 mg, 2.0 mmol) and CH_3CN (7.0 mL) were added sequentially under a nitrogen atmosphere. The reaction was complete after being stirred at 50 °C for 2 h as monitored by TLC (eluent: petroleum ether/ethyl acetate=

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Scheme 2. The Synthesis of (1*R*,3*R*,7*S*)- and (1*S*,3*S*,7*R*)-**3l** with a Chiral Center at the Propargylic Position.



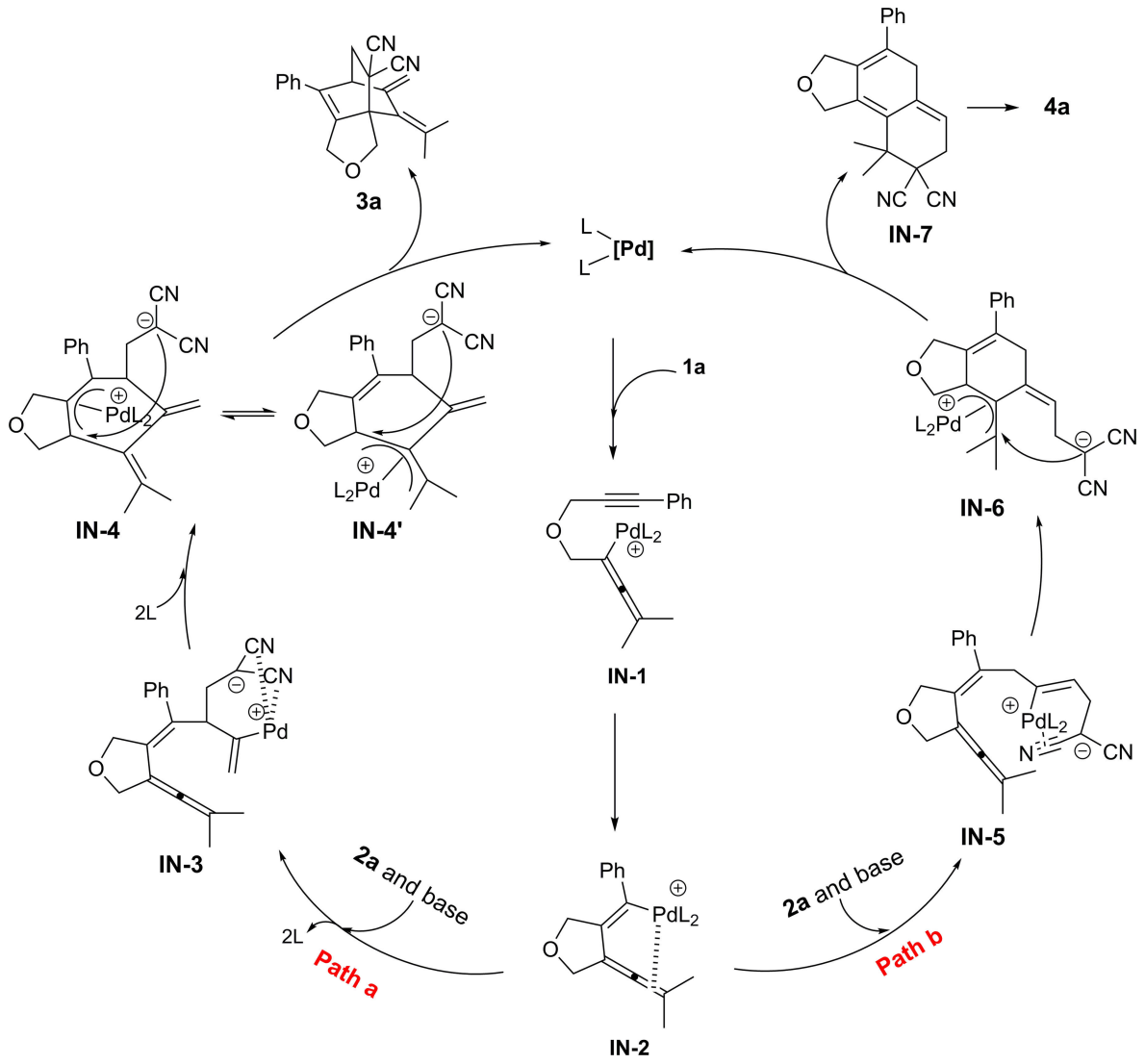
1	2	t/h	NMR Yield of **3** (%)	Recovery of **1** (%)
X = O, R^1 = Me	**2a**	1.5	61	-
X = CH_2, R^1 = Me	**2a**	17	Complicated	-
X = NTs, R^1 =Me	**2a**	22	-	71
X = S, R^1 = Et	**2a**	15.5	-	77
X = O, R^1 = Me	**2b**	5.5	Complicated	-

Scheme 3. The Unique Role of the Oxygen Tether in **1** and Two Cyano Groups in **2.**

5/1). After cooling to room temperature, the resulting mixture was filtered through a short column of silica gel eluted with ethyl acetate (10 mL×3). The reaction mixture was concentrated to afford a mixture of **3a**, **4a**, and **5a**. The NMR yields were determined by ^{1}H NMR analysis of the crude product with mesitylene (46 μL) as the internal standard (**3a** (64%), **4a** (2%), and **5a** (1%)). Chromatography on silica gel (eluent: petroleum ether/$CHCl_3$/CH_3CN = 14/1/0.3) afforded impure **3a**, which was further purified by recrystallization from DCM (2 mL) and petroleum ether (6 mL) to afford the pure **3a** (166.0 mg, 51%) as a solid: M.P. 138.1–138.6 °C (petroleum/DCM); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.46–7.28 (m, 3H, Ar–H), 7.24–7.16 (m, 2H, Ar–H), 5.11 (s, 1H, one proton of =CH_2), 4.87 (d, J = 10.5 Hz, 1H, one proton of OCH_2), 4.83 (s, 1H, one proton of =CH_2), 4.74 (d, J = 13.5 Hz, 1H, one proton of OCH_2), 4.51 (d, J = 13.8 Hz, 1H, one proton of OCH_2), 4.40 (d, J = 10.2 Hz, 1H, one proton of OCH_2), 3.82–3.77 (bs, 1H, CH), 2.57 (dd, J_1 = 13.5 Hz, J_2 = 1.4 Hz, 1H, one proton of CH_2), 2.48 (dd, J_1 = 13.8 Hz, J_2 = 3.3 Hz, 1H, one proton of CH_2), 2.06 (s, 3H, CH_3), 2.00 (s, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ 148.3, 139.5, 138.8, 137.1, 135.4, 128.8, 128.3, 126.4, 124.6, 116.2, 115.6, 110.2, 73.3, 68.0, 61.0, 47.3, 39.1, 38.8, 26.2, 25.6; IR (neat, cm^{-1}) 3084, 3055, 3024, 2968, 2926, 2867, 2840, 2245, 1634, 1617, 1599, 1497, 1444, 1379, 1363, 1345, 1272, 1227, 1206, 1162, 1129, 1087, 1063, 1043, 1001; MS (EI, 70 eV) *m/z* (%) 329 (M^+ +1, 22.71), 328 (M^+, 100), 327 (M^+ −1, 70.20); Anal. Calcd for $C_{22}H_{20}N_2O$: C 80.46; H 6.14; N 8.53; Found: C 80.57; H 6.20; N 8.53.

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Scheme 4. Proposed mechanism for the formation of **3a** and **4a.**

Acknowledgements

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[8] **3a**: $C_{22}H_{20}N_2O$, MW = 328.40, orthorhombic, space group P b c a, final R indices I > 2sigma(I), R1 = 0.0411, wR2 = 0.0937; R indices (all data), R1 = 0.0535, wR2 = 0.1010; a = 7.7892(5) Å, b = 14.6752(7) Å, c = 30.3163(13) Å, α = 90.00°, β = 90.00°, γ = 90.00°, V = 3465.4(3) Å^3, T = 293(2) K, Z = 8, reflections collected/unique 21452/3167 (Rint = 0.0426), number of observations [> 2σ(I)]: 2596, parameters: 228. CCDC 1542004 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

[9] (*1R*, *3R*, *7S*)-**3l**: $C_{23}H_{22}N_2O$, MW = 342.42, monoclinic, space group P 1 21 1, final R indices I > 2\s(I), R1 = 0.0408, wR2 = 0.1013; R indices (all data), R1 = 0.0454, wR2 = 0.1070; a = 13.8440(3) Å, b = 8.4708(2) Å, c = 15.9727(3) Å, α = 90.00°, β = 91.0651(19)°, γ = 90.00°, V = 1872.79(7) Å^3, T = 293(2) K, Z = 4, reflections collected/unique 20497/6434 (Rint = 0.0497), number of observations [> 2σ(I)]: 5848, parameters: 475. CCDC 1541998 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif. (*1S*, *3S*, *7R*)-**3l**: $C_{23}H_{22}N_2O$, MW = 342.42, monoclinic, space group P 1 21 1, final R indices I > 2\s(I), R1 = 0.0412, wR2 = 0.1035; R indices (all data), R1 = 0.0492, wR2 = 0.1141; a = 13.8462(3) Å, b = 8.4832(2) Å, c = 15.9886(4) Å, α = 90.00°, β = 91.108(2)°, γ = 90.00°, V = 1877.67(9) Å^3, T = 293(2) K, Z = 4, reflections collected/unique 12553/6525 (Rint = 0.0502), number of observations [> 2σ(I)]: 5638, parameters: 476. CCDC 1541997 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

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